

Modbus FCs & Registers

Protocol	FCs	Registers	Address (Decimal)	
Modbus TCP	FC04 Read	Analog Input Registers	Temperature (Read Only)	Port 1 : 001 Port 2 : 201 Port 3 : 401 Port 4 : 601
			Buffer Data Length (Read Only)	Port 1 : 002 Port 2 : 202 Port 3 : 402 Port 4 : 602
			16-Bit Raw Data (Ring Buffer) (Read Only) X, Y, Z X, Y, Z Total 41 data sets (Without Offset) Value = Register Value ÷ Sensitivity	Port 1 : 002 Port 2 : 202 Port 3 : 402 Port 4 : 602
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	Output Data Rate (ODR) (Read / Write)	001
			Offset (X / Y / Z) (Read / Write)	Port 1 : 002 / 003 / 004 Port 2 : 202 / 203 / 204 Port 3 : 402 / 403 / 404 Port 4 : 602 / 603 / 604
			Upper Limit (X / Y / Z) (Read / Write)	Port 1 : 005 / 007 / 009 Port 2 : 205 / 207 / 209 Port 3 : 405 / 407 / 409 Port 4 : 605 / 607 / 609
			Lower Limit (X / Y / Z) (Read / Write)	Port 1 : 006 / 008 / 010 Port 2 : 206 / 208 / 210 Port 3 : 406 / 408 / 410 Port 4 : 606 / 608 / 610
			Alarm State (Read / Write) 0 : Normal 1 : Alarm	011
			Watch Dog (Read / Write) N : N Seconds -1 : Disable 0 : Alarm	012

			Sensor State (Read Only) Failure (1) or Normal (0) : Bit 7 ~ Bit 4 = Port 4 ~ Port 1 Connected (1) or Not (0) : Bit 3 ~ Bit 0 = Port 4 ~ Port 1	013
			Communication Mode (Read / Write) 0 : Ethernet (Modbus TCP) 1 : Ethernet (UDP) 2 : WiFi STA 3 : WiFi AP 4 : WiFi Settings	014
			I/O Mode (Read / Write) 0 : I/O Off 1 : Alarm single port 2 : Alarm all ports	015
			Sensor Type (Read / Write) 1 : LNAX301 (2G) 2 : LNAX301 (2.5G) 3 : LNAX301 (4G) 4 : LNAX301 (8G) 5 : LNAX301 (10G) 6 : KAX301 (8G)	016
			IP Address (Read / Write) 192 . 168 . [I] . [II]	I : 017 II : 018
			Gateway Address (Read / Write) 192 . 168 . [I] . [II]	I : 117 II : 118
			Device ID	019
			Temperature (Read Only) Value = Register Value ÷ 112.9761 - 7.963	Port 1 : 020 Port 2 : 220 Port 3 : 420 Port 4 : 620
			Bulk Transfer Mode (Write Only)	Port 1 : 021 Port 2 : 221 Port 3 : 421 Port 4 : 621

			Clear Buffer (Write Only) Bit 7 ~ Bit 4 = 0X00 (N/A) Clear (1) or Don't (0) : Bit 3 ~ Bit 0 = Port 4 ~ Port 1	022
			I/O Input State (Read Only) Bit 7 ~ Bit 4 = 0X00 (N/A) High (1) or Low (0) : Bit 3 ~ Bit 0 = DI 4 ~ DI 1	023
			I/O Output State (Read / Write) Bit 7 ~ Bit 4 = 0X00 (N/A) High (1) or Low (0) : Bit 3 ~ Bit 0 = DO 4 ~ DO 1	024
			RMS (X / Y / Z) (Read Only) Value = Register Value ÷ 1000	Port 1 : 025 / 026 / 027 Port 2 : 225 / 226 / 227 Port 3 : 425 / 426 / 427 Port 4 : 625 / 626 / 627
			Peak (X / Y / Z) (Read Only) Value = Register Value ÷ 1000	Port 1 : 028 / 029 / 030 Port 2 : 228 / 229 / 230 Port 3 : 428 / 429 / 430 Port 4 : 628 / 629 / 630
			Crest Factor (X / Y / Z) (Read Only) Value = Register Value ÷ 1000	Port 1 : 031 / 032 / 033 Port 2 : 231 / 232 / 233 Port 3 : 431 / 432 / 433 Port 4 : 631 / 632 / 633
			Kurtosis (X / Y / Z) (Read Only) Value = Register Value ÷ 1000	Port 1 : 034 / 035 / 036 Port 2 : 234 / 235 / 236 Port 3 : 434 / 435 / 436 Port 4 : 634 / 635 / 636
			Peak Frequency (X / Y / Z) (Read Only) Value = Register Value ÷ 10	Port 1 : 037 / 038 / 039 Port 2 : 237 / 238 / 239 Port 3 : 437 / 438 / 439 Port 4 : 637 / 638 / 639
			Wavelet Packets Node 1 ~ 32 (X / Y / Z) (Read Only) Value = Register Value ÷ 1000	Port 1 : 040 ~ 071 / 072 ~ 103 / 104 ~ 135 Port 2 : 240 ~ 271 / 272 ~ 303 / 304 ~ 335 Port 3 : 440 ~ 471 / 472 ~ 503 / 504 ~ 535 Port 4 : 640 ~ 671 / 672 ~ 703 / 704 ~ 735

			Data Mode 0 : 16-Bit Raw Signal 1 : Time Domain Parameter 2 : Frequency Domain 3 : Wavelet Packets (Peak) 4 : Wavelet Packets (RMS) 5 : Wavelet Packets (Variance) 6 : Wavelet Packets (Crest Factor)	1000
			IP Address (Read / Write) [0]. [1] . [xxx] . [yyy]	[0] : 2000 [1] : 2001
			RSSI	2002
			REBOOT WIFI BOX Write Any Value (0~65535)	2004
			前後封包時間差距 (Read)	2005
			Reset WIFI Module (Write)	2006
			PING 成功次數	2007
			PING 失敗次數	2008
			切換同步(寫入 0)/非同步(寫入 1) 模式 預設是同步模式 同步模式 NetworkCommunication() 主迴圈做 (速度慢) 非步模式 NetworkCommunication() 計時函式 做(速度快) (Write/Read)	2009
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	Gateway Address (Read / Write) [0]. [1] . [2] . [3]	[0] : 2010 [1] : 2011 [2] : 2012 [3] : 2013
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	Netmask (Read / Write) [0]. [1] . [2] . [3]	[0] : 2014 [1] : 2015 [2] : 2016 [3] : 2017
Modbus TCP &	FC03 Read & FC06	Analog Output Holding Registers	Write Preset Write Any Value (0~65535)	2018

Modbus RTU	Write Single			
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	Hex Total Lines for OTA	2019
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	PSRAM SIZE MB	2020
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	SDCARD exist? 1:存在 0: 不存在	2021
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	OTA HEX RX LINES	2022
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	Sensor Reconnect 寫奇數值	2023
Modbus TCP & Modbus RTU	FC03 Read & FC06 Write Single	Analog Output Holding Registers	寫 1: 進入錄 raw data mode 寫 0: 離開錄 raw data mode	2024